Introduction to Software Engineering

Software Requirements Specification

The student team is required to complete the **Software Requirements Specification** documentation for the assigned course project, following the attached template.



Software Engineering Department
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Table of Contents

Objectives **1**
1 Member Contribution Assessment **2**
2 Introduction **5**
2.1 Purpose and Scope 5
Purpose 5
Product Scope 5
2.2 Glossary 6
3 Problem Statement **8**
4 Stakeholders **10**
5 Requirements **11**
5.1 Functional Requirements 11
5.2 Non-Functional Requirements 15
5.3 Domain Requirements and Business Rules 16
5.4 Acceptance Criteria 18
6 Use Case Model **20**
6.1 Use Case Diagram 20
6.2 Actors 21
6.3 Use Case List 21
7 Use Case Specifications **23**
7.1 Use Case UC-01 23
7.2 Use Case UC-02 24
7.3 Use Case UC-03 25
7.4 Use Case UC-04 26
7.5 Use Case UC-05 27
7.6 Use Case UC-06 28
7.7 Use Case UC-07 29
7.8 Use Case UC-08 30
7.9 Use Case UC-09 31
7.10 Use Case UC-10 33
7.11 Use Case UC-11 34
7.12 Use Case UC-12 35
7.13 Use Case UC-13 37
7.14 Use Case UC-14 38
8 Prototype / Mockups **41**
9 Requirements Traceability Matrix **42**
10 Requirements Review **46**
11 AI Usage Log **47**

Software Requirements Specification

Objectives

This document focus on the following topics:

- ✔ Completing the Software Requirements Specification (SRS) document with the following sections:
  - Problem Statement
  - Stakeholders
  - Functional, non-functional and domain requirements
  - Use case model and use case specifications
  - Prototype / mockups
  - Traceability and requirements review
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Software Requirements Specification (SRS) document.

1 Member Contribution Assessment

ID	Name	Role(s)	Contribution (%)	Evidence
24127455	Lại Phan Nhật Nam	Team Lead	20	7.1 Weekly status meeting 7.2 Weekly status report > Commit > Item 2.3.1 Use cases: Use case for the must-have requirements > Commit > Item 1 - Item 2 - Item 3 - Item 4 PA2: Complete section 4, 5.1 > Commit > Item (Item of 5.1 is the same as 2.3.1) PA2: Complete section 6.2 > Commit > Item PA2: Complete section 9 > Commit > Item PA2: Complete section 1, 10 > Commit > Item
24127295	Phạm Hoàng Khôi Nguyên	Business Analyst	20	2.1 Vision, refined for PA2 > Pull request > Item 2.2.1 SRS: Must-have functional requirements > Pull request > Item

				2.2.2 SRS: Other functional requirements > Pull request (Item of 2.2.2 is the same as 2,2,1) 2.2.3 SRS: Non-functional requirements (Measureable) > Pull request (Item of 2.2.3 is the same as 2,2,1) PA2: Complete section 7
24127459	Trần Văn Nam	Designer	20	2.4.3 Mockups: Expense screen > Pull request > Item 2.4.2 Mockups: Group and invite screens (Include QR join) > Pull request > Item 2.4.1 Mockups: Account screen > Pull request > Item PA2: Manage AI usage log PA2: Complete section 6.1, 8 2.4.4 Mockups: Debt screen > Pull request > Item PA2: Complete section 2, 11
24127308	Nguyễn Hoàng Tiến	Developer	20	Help with SRS: - 2.2.1 - 2.2.2 - 2.2.3 2.3.2 Use cases: Use case for other requirements > Pull request > Item

				PA2: Complete section 5.2 PA2: Complete section 3, 5.3 > Pull request > Pull request > Item 1 - Item 2
24127409	Ngô Phúc Khang	Tester	20	Help with mockups: - Expense screen - Group & invite screen - Account screen PA2: Complete section 5.4 PA2: Complete section 6.3 4.1 Test plan > Pull request > Item

2 Introduction

2.1 Purpose and Scope

Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements of SplitNest, a web application for managing shared expenses and repayments.

The purpose of this document is to describe the system's expected behavior, scope, users, and constraints in a clear and structured manner. It provides a common reference for the development team, testers, designers, instructors, and other stakeholders throughout the requirements, design, implementation, and testing phases.

The document also establishes the basis for use case modeling, prototype development, and requirements verification.

Product Scope

SplitNest is a Vietnamese-first web application designed for friends, families, and group-fund treasurers who need to manage shared expenses and track repayments.

The system allows users to register and sign in, create or join private groups, record shared expenses, calculate individual shares, and track outstanding debts. It also supports a repayment confirmation workflow in which the debtor reports a repayment and the receiver confirms or rejects it.

In Scope:

- User registration and sign-in.

- Saving personal bank details as text, including bank name, account number, and account holder name.
- Creating private groups and inviting registered users through join links or QR codes.
- Joining groups and viewing group membership.
- Recording shared expenses with one payer and selected participants.
- Dividing expenses equally in whole Vietnamese Dong (VND), following the defined remainder distribution rule.
- Tracking individual debts and outstanding amounts.
- Reporting repayments and allowing receivers to confirm or reject them.
- Editing or deleting expenses under the defined business rules.

Out of Scope:

- Automatic bank transfers, bank API integration, and automatic payment verification.
- Support for currencies other than VND.
- Native Android and iOS applications.
- Email and push notifications for unpaid debts.
- In-app chatting and debt-related notes.
- Managing real money in a shared group fund.
- Repaying debts through smaller installments.

2.2 Glossary

Term	Definition
Expense	A shared cost recorded in a group, with one payer, one or more participants, and a positive whole-VND amount.
Payer	The group member who paid an expense. For each obligation created by that expense, the payer is the creditor.
Participant	A group member who shares the cost of an expense. The payer may also be a participant.

Share	The whole-VND amount allocated to one participant. Shares of an expense sum exactly to its total, with any remainder assigned in member-ID order.
Repayment obligation (debt)	The amount a participant owes the payer for their share of one expense, linked to that source expense. No obligation is created for the payer's own share.
Debtor	The member who owes an obligation. Only the debtor can report repayment.
Creditor	The member owed an obligation (the expense's payer). Only the creditor can confirm or reject a repayment report.
Outstanding	Obligation status: no repayment has been reported, or the last report was rejected.
Pending Confirmation	Obligation status: the debtor has reported full repayment and the creditor has not yet decided. The obligation is still unsettled.
Confirmed	Obligation status: the creditor has confirmed receipt. The obligation is settled and cannot be settled again.
Unsettled debt	An obligation that is Outstanding or Pending Confirmation. This is the condition that blocks leaving or removal (FR-40, FR-41).
Repayment report	A debtor's statement that they repaid the full amount of one obligation outside SplitNest. It is not proof of payment and does not settle the obligation.
Rejected report	A repayment report the creditor declined, typically because they did not receive the payment. The obligation returns to Outstanding. The report stays on record and still prevents the source expense from being edited or deleted (FR-28).
Offsetting	Netting obligations in opposite directions between two members. SplitNest does not do this.
Organizer	The group's creator. They can get the invitation link and QR code and, where supported, remove members.
Invitation link	A link, also shown as a QR code, that lets a signed-in user join a group.
VND	Vietnamese Dong, the only supported currency, always used in whole amounts.

3 Problem Statement

Business context and current problems

Groups of friends, families, and small group-fund participants regularly share expenses for meals, groceries, and activities. Different members may pay for different expenses, while only selected members participate in each purchase. Members therefore need a common record of what was purchased, who paid, how the cost was divided, and which repayments remain outstanding.

When these records are maintained through chat messages or spreadsheets, expense details and repayment updates can become scattered or inconsistent. Uneven division of whole-VND amounts can also produce rounding differences unless the group follows a consistent allocation rule. A further problem is the distinction between reporting a repayment and confirming receipt. Treating a reported payment as immediately settled can create disagreements and inaccurate outstanding totals. Changing an expense after repayment activity has begun can also make the original debt and its settlement records inconsistent.

Goals and proposed approach

SplitNest aims to provide a shared, understandable record of group expenses and repayment status. Users can create accounts, create or join private groups, record expenses, and view the amounts they owe or are owed. Each expense has one payer and selected participants, with costs divided equally in whole VND using a deterministic remainder rule.

Each repayment obligation remains linked to its source expense. Debt summaries are calculated from these individual obligations without automatically offsetting debts in opposite directions. A debtor reports full repayment of a specific obligation made outside the application. The creditor then confirms or rejects the report; pending reports remain unsettled until confirmation.

Restrictions on expense editing and deletion protect records once repayment reporting has begun.

The intended benefit is to reduce manual reconciliation and uncertainty about outstanding debts while giving group members and organizers access to consistent information.

Operating environment

SplitNest will operate as an internet-connected web application for desktop and mobile browsers. The core features will be tested on desktop Chrome, Edge, Firefox, and one mobile browser. The planned implementation uses React and JavaScript for the frontend, Flask and Python for the backend, and MongoDB Atlas for persistent storage. Deployment is planned on Vercel and Render, using free-tier services where possible. Hosting limitations, including backend cold-start delays, form part of the operating constraints.

Design and implementation constraints

The initial release prioritizes accounts and bank details, private groups and invitations, shared expense recording, and debt tracking and settlement. A five-person student team is scheduled to implement the application during weeks 8–10, so optional features must remain subordinate to completing and verifying these core workflows.

The application supports VND only, equal splitting, and full repayment of individual obligations. Group data is restricted to group members. Users may optionally store bank details as text, without connecting a bank account. VietQR payment instructions and chronological history are optional enhancements. Out-of-scope functions and detailed requirements are specified in Section 2.1 and Section 5, respectively.

4 Stakeholders

#	Stakeholder	Description	Interests / concerns
1	Project sponsor (Instructor / TA)	Academic stakeholders who set the project scope, review each PA and evaluate the final system.	A clear problem, traceable requirements, a feasible scope, and a working demonstrable system by week 11.
2	Group organizer / treasurer	A registered user who creates a group, invites members, and may also take part in expenses.	Simple group setup, easy invitations (link/QR), and an overview of who still owes money.
3	Group member	A registered user who joins groups, records expenses, and may owe or be owed money.	Accurate balances, fair whole-VND splits, access limited to group members, and clear debt statuses.
4	Debtor (situational role)	A member who owes another member for an expense and reports repayment made outside SplitNest.	Easy repayment reporting, a clear view of what is still owed, and no wrongly recorded debts.
5	Creditor (situational role)	The member who paid for an expense and confirms or rejects repayment reports.	Control over confirmation, protection against false reports, and no double settlement.
6	Project team	Five students who analyse, design, build, test and deliver the system.	Requirements that are clear, testable and achievable in weeks 8-10 with free-tier tools.
7	Hosting and service providers (Vercel, Render, MongoDB Atlas)	Third-party platforms hosting the frontend, backend and database.	Usage within free-tier limits. This constrains availability and performance.

5 Requirements

5.1 Functional Requirements

The sources are derived from section 2.4 of PA1.

ID	Requirement	Feature	Priority	Source
FR-01	The system shall allow users to create an account using valid registration information.	F01	Must	Group member
FR-02	The system shall reject registration if the user identifier is already in use or required information is invalid.	F01	Must	Group member
FR-03	The system shall allow users to sign in with valid credentials.	F01	Must	Group member
FR-04	The system shall reject sign-in attempts with incorrect credentials.	F01	Must	Group member
FR-05	The system shall allow users to optionally save their bank name, account number and account holder name as text.	F01	Must	Group member
FR-06	The system shall allow users to view and edit their own bank details.	F01	Must	Group member
FR-07	The system shall allow a signed-in user to sign out.	F01	Must	Group member
FR-08	The system shall end the user's session on sign-out, so that protected group data cannot be accessed until the user signs in again.	F01	Must	Group member
FR-09	The system shall allow signed-in users to create private groups for managing shared expenses.	F02	Must	Group organizer / treasurer
FR-10	The system shall record the group creator as a member of that group.	F02	Must	Group organizer / treasurer
FR-11	The system shall allow the group organizer to get the group's invitation link.	F02	Must	Group organizer / treasurer
FR-12	The system shall display a QR code containing the group invitation link.	F02	Must	Group organizer / treasurer

FR-13	The system shall allow signed-in users to join a group using a valid invitation link.	F02	Must	Group organizer / treasurer
FR-14	The system shall reject invalid or expired invitation links without adding the user to a group.	F02	Must	Group organizer / treasurer
FR-15	The system shall allow group members to view the list of members in their group.	F02	Must	Group member
FR-16	The system shall prevent non-members from accessing group data or adding expenses to the group.	F02	Must	Group member
FR-17	The system shall not add a duplicate member when an existing member reopens the invitation link.	F02	Must	Group organizer / treasurer
FR-18	The system shall allow a group member to record an expense with a positive whole-VND amount, one payer and at least one participant.	F03	Must	Group member
FR-19	The system shall require the payer and all participants to belong to the group.	F03	Must	Group member
FR-20	The system shall reject expenses with a zero, negative, non-integer or otherwise invalid amount.	F03	Must	Group member
FR-21	The system shall split an expense evenly in whole VND and assign any remainder in a fixed order (by member ID).	F03	Must	Project team
FR-22	The system shall ensure the sum of all participants' shares equals the total expense exactly.	F03	Must	Project team
FR-23	The system shall create a repayment obligation, linked to its source expense, for each participant other than the payer.	F03	Must	Group member
FR-24	The system shall not create an obligation for a person to repay themselves.	F03	Must	Group member
FR-25	The system shall allow group members to view expense details, payer, participants and allocated amounts.	F03	Must	Group member
FR-26	The system shall allow only the member who created an expense to edit or delete it, provided no related repayment report has been submitted.	F03	Must	Project team

FR-27	The system shall recalculate or remove related debts when an expense is edited or deleted.	F03	Must	Project team
FR-28	The system shall reject editing or deleting an expense after any related repayment report exists, even a rejected one.	F03	Must	Project team
FR-29	The system shall show each amount a user owes or is owed within the group.	F04	Must	Group member
FR-30	The system shall calculate total debts from individual obligations without automatically offsetting debts in opposite directions.	F04	Must	Project team
FR-31	The system shall allow only the debtor to report full repayment of their outstanding obligation.	F04	Must	Debtor / creditor
FR-32	The system shall change a debt's status to Pending Confirmation, and keep it unsettled, when repayment is reported.	F04	Must	Debtor / creditor
FR-33	The system shall allow only the creditor to confirm a repayment, changing the debt status to Confirmed.	F04	Must	Debtor / creditor
FR-34	The system shall allow the creditor to reject a repayment, returning the debt to Outstanding.	F04	Must	Debtor / creditor
FR-35	The system shall reject attempts to confirm or reject a repayment by anyone other than the creditor.	F04	Must	Debtor / creditor
FR-36	The system shall not accept partial repayment reports in the first release.	F04	Must	Project team
FR-37	The system shall not allow an already Confirmed debt to be settled again.	F04	Must	Debtor / creditor
FR-38	The system shall not allow multiple simultaneous Pending Confirmation requests for the same debt.	F04	Must	Debtor / creditor
FR-39	The system shall update the displayed debt totals when a repayment report is confirmed or rejected.	F04	Must	Debtor / creditor
FR-40	The system shall allow a group member other than the organizer to leave a group only if the member has no unsettled debts in that group, either as debtor or as creditor.	F02	Should	Group member

FR-41	The system shall allow the group organizer to remove a member from the group only if that member has no unsettled debts in that group, either as debtor or as creditor.	F02	Should	Group organizer / treasurer
FR-42	The system shall reject a leave or removal request when the member has any unsettled debt in the group, and shall tell the user why.	F02	Should	Group member
FR-43	The system shall deny a member who has left or been removed access to the group's data and shall exclude them from selection as payer or participant in new expenses.	F02	Should	Group member
FR-44	The system shall retain a former member's confirmed debts and past expenses in the group's records.	F02	Should	Project team
FR-45	The system shall not allow the group organizer to leave the group or be removed from it in the first release.	F02	Should	Project team
FR-46	The system shall allow only the group organizer to remove members; other members shall not be able to remove anyone.	F02	Should	Group organizer / treasurer
FR-47	The system shall allow a former member to join the group again through a valid invitation link as a new member.	F02	Should	Group organizer / treasurer
FR-48	The system shall generate a VietQR payment instruction from the recipient's saved bank details and the repayment amount, without initiating a transfer.	F05	Should	Group member
FR-49	The system shall display the generated VietQR for payment through an external banking application.	F05	Should	Group member
FR-50	The system shall display recorded group expenses in chronological order.	F06	Could	Group member
FR-51	The system shall display repayment, confirmation and rejection activity in chronological order.	F06	Could	Group member
FR-52	The system shall support notifications for unpaid debts in a future release.	F07	Won't	Project team
FR-53	The system shall support messages and debt-related notes in a future release.	F08	Won't	Project team

5.2 Non-Functional Requirements

ID	Category	Requirement (measurable)	How it will be verified
NFR-01	Performance	95% of expense and debt queries shall finish within 3 seconds, with 10 simulated concurrent users, up to 100 group expenses, and the backend already running.	Load testing; measure P95
NFR-02	Security	Reject 100% of unauthorized requests for group data without returning protected information.	API authorization tests
NFR-03	Security	Do not store passwords in plaintext in the database or logs.	Check database and logs
NFR-04	Security	All deployed public pages and APIs shall use HTTPS.	Check deployed protocols
NFR-05	Data integrity	100% of tested expense splits shall preserve the total amount and use only whole-VND values.	Unit tests for the splitting algorithm
NFR-06	Concurrency	20 nearly simultaneous confirmation requests shall not result in multiple successful settlements of the same debt.	Concurrent integration tests
NFR-07	Reliability	Successfully saved data shall remain intact after refreshing, signing out and back in, and restarting the application across 20 test runs.	Persistence tests
NFR-08	Usability	At least 4 out of 5 first-time users shall complete the create/join group, record expenses, and view debt workflow within 5 minutes without assistance.	Usability testing

NFR-09	Compatibility	All four Must-have features shall pass smoke tests on desktop Chrome, Edge, Firefox, and one mobile browser.	Browser testing
NFR-10	Accessibility	All F01–F04 controls shall have accessible names and support keyboard use; regular text on main screens shall have a contrast ratio of at least 4.5:1.	Accessibility audit
NFR-11	Maintainability	Every code-changing pull request shall run frontend and backend tests and shall not be merged if a required job fails.	GitHub Actions/branch protection
NFR-12	Privacy	Group, expense and debt list APIs shall not include users' saved bank details. A creditor's bank details shall be returned only to a debtor of that creditor when the debtor requests payment information for an outstanding obligation (including VietQR generation), and to no one else.	API response inspection

5.3 Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	The system shall support VND only. Expense amounts and allocated shares shall be represented in the whole VND.	PA1 2.1, 2.4
BR-02	The system shall record expenses and repayment information without holding money, initiating bank transfers, or automatically verifying receipt of payment. Actual transfers shall take place outside the application.	PA1 2.1. 2.4
BR-03	The creator of a group shall automatically become a member of that group. Reopening an invitation link shall not create duplicate membership for an existing member.	FR-10, FR-17
BR-04	Only members of a group shall be permitted to access its protected data or record expenses within it.	FR-16

BR-05	Each expense shall have a positive whole-VND amount, exactly one payer, and at least one participant sharing the cost. The payer and all participants shall belong to the same group as the expense.	FR-18, FR-19, FR-20
BR-06	Each expense shall be split evenly among its selected participants in whole VND. Any remainder shall be distributed according to a fixed participant ordering, and the allocated shares shall sum exactly to the expense amount.	FR-21, FR-22
BR-07	Each participant other than the payer shall have an individual repayment obligation to the payer for their allocated share. No self-repayment obligation shall be created.	FR-23, FR-24
BR-08	Only the member who created an expense shall be permitted to edit or delete it, and only before any related repayment report has been submitted. Submission of any related report shall permanently prevent editing or deletion, even if that report is later rejected.	FR-26, FR-27
BR-09	Debt totals shall be calculated from individual repayment obligations. Obligations in opposite directions shall not be automatically offset against each other.	FR-29, FR-30
BR-10	Only the debtor of an outstanding obligation shall be permitted to report its repayment. Each report shall cover the full obligation amount; partial repayment reports shall not be accepted in the first release.	FR-31, FR-36
BR-11	Reporting repayment shall change the obligation status from Outstanding to Pending Confirmation. A pending report shall not count as a settled obligation.	FR-32
BR-12	Only the creditor of the obligation shall be permitted to confirm or reject a repayment report. Confirmation shall change the obligation status to Confirmed; rejection shall return it to Outstanding.	FR-33, FR-34, FR-35
BR-13	An obligation shall have no more than one pending repayment report at a time. A Confirmed obligation shall not be settled again.	FR-37, FR-38
BR-14	Displayed debt totals shall reflect repayment decisions. Confirmed obligations shall be excluded from outstanding debt totals; rejected obligations shall remain outstanding.	FR-39
BR-15	Saving bank details shall be optional. Saved bank details shall consist of the bank name, account number, and account holder name, stored as text. Users shall be permitted to view and edit their own saved bank details.	PA1 2.4; FR-05
BR-16	VietQR payment instructions shall use the recipient's saved bank details and the repayment amount. Generating or displaying these instructions shall not	FR-48, FR-49; PA1 2.4

	itself constitute a repayment report or confirmation and shall not settle the obligation.	
BR-17	Expense history shall display recorded group expenses in chronological order. Repayment activity shall display repayment reports and their confirmation or rejection events in chronological order, preserving the distinction between reporting a payment and confirming its receipt.	FR-50-51; PA1 2.4

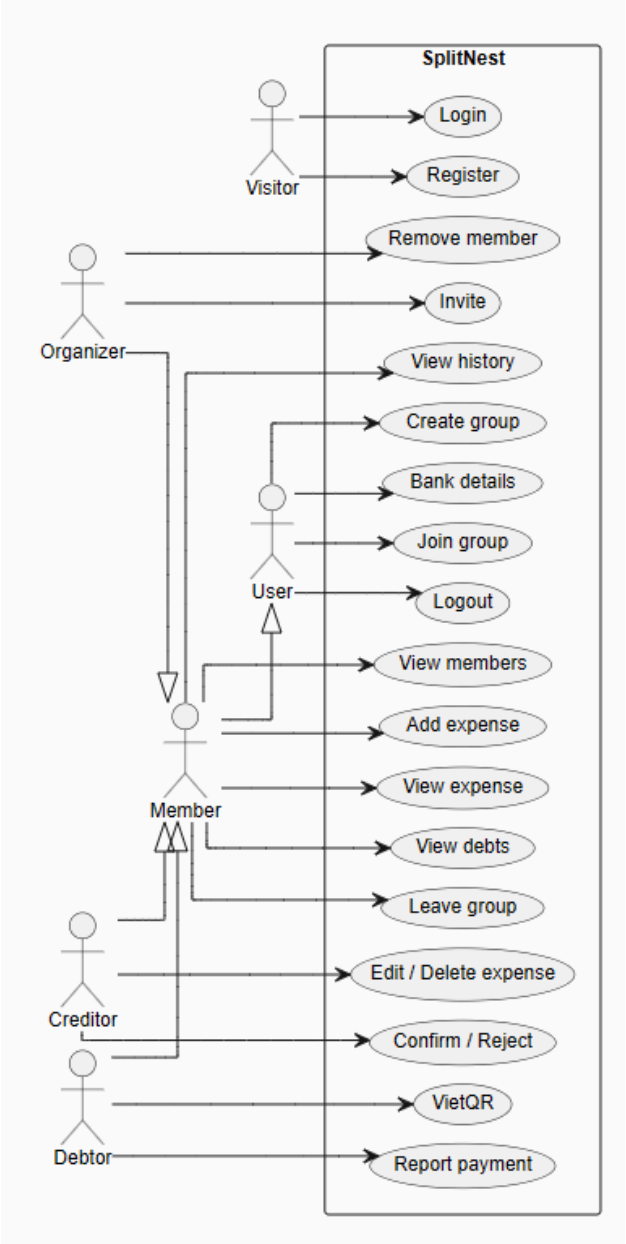
5.4 Acceptance Criteria

ID	Scenario	Given	When	Then
AC-01.1	Successful Registration	An unregistered visitor is on the registration page.	They submit a valid email and a password meeting security criteria, with or without optional bank details (bank name, account number, account holder name).	The system creates the account, securely hashes the password, stores any bank details provided as text, and signs the user in.
AC-01.2	Duplicate Email Rejection	An email address that is already registered in the system.	A visitor attempts to register using that same email address.	The system rejects the registration and displays an error message stating "Email is already registered".
AC-01.3	Update Bank Details	An authenticated user is accessing their account profile settings.	They edit any of the bank text fields (e.g., account number) and click "Save Changes".	The system updates the profile records without initiating any external bank verification and confirms the update.
AC-02.1	Group Creation	An authenticated user	They provide a non-empty group name and submit the creation form.	The system creates the new group, designates the user as the organizer, and generates a unique invitation link/QR code.
AC-02.2	Join Group via Invitation Link/QR code	An authenticated user who is not currently a member of Group A.	They navigate to Group A via a valid invite link/QR code and confirm to join.	They are added to Group A's member list and gain access to view its shared expenses and balances.
AC-02.3	Invalid Invitation Link/QR code	An authenticated user.	They navigate to a group A via a invalid, malformed or expired group invite link/QR code	The system rejects the request, displays an error message ("Invalid Link/QR code"), and does not grant group access.
AC-03.1	Equal Split	A group containing 3	User A logs an expense of	The system records the

	Calculation	members (A, B, and C).	300,000 VND, identifies as the payer, and selects all 3 members to split equally.	expense, assigns a 100,000 VND share to each participant, and updates the group ledger.
AC-03.2	Rounding & Integer Allocation (NFR-05)	An expense of 100,000 VND to be split equally among 3 members.	The split calculation is executed.	The system allocates 33,334 VND to A and 33,333 VND to each of B and C. All shares are whole-VND amounts, the difference between any two shares does not exceed 1 VND, and their sum equals exactly 100,000 VND.
AC-03.3	Invalid Expense Input Validation	An authenticated user filling out the expense form.	They enter a zero, negative, or non-integer amount, or submit without selecting at least one participant.	The system blocks submission, highlights the invalid fields, and displays appropriate validation alerts.
AC-04.1	View Individual Obligations	A group with recorded transactions where User A owes User B 50,000 VND.	User A views the debt watching page.	The system displays a summary stating User A owes User B exactly 50,000 VND.
AC-04.2	Mark Debt as Paid	An unsettled debt of 50,000 VND between debtor User A and creditor User B.	User A clicks "Mark as paid".	The debt status changes to "*Pending Confirmation*", and a settlement confirmation request appears for User B.
AC-04.3	Confirm Payment Receipt	A pending settlement between User A and User B with status "*Pending Confirmation*".	Creditor User B clicks "Confirm Received".	The obligation status changes to "*Confirmed*", it is excluded from outstanding totals, and the decision is recorded. The system cannot settle the same obligation again.
AC-04.4	Reject Settlement	A settlement in "*Pending Confirmation*" status where User B has not received the actual funds.	User B clicks "*Reject / Not Received*".	The debt status reverts to "*Outstanding*", the balance remains outstanding.

6 Use Case Model

6.1 Use Case Diagram



6.2 Actors

Actor	Description
Visitor	A person who is not signed in. They can register and log in.
Signed-in user	A registered person with an active session. They can manage their own bank details, create groups, join groups through an invitation link and sign out.
Group member	A signed-in user who belongs to a group. They can view members, record and view expenses, and view debts within that group. A member cannot see data from groups they don't belong to.
Group organizer	The member who created the group. They can do everything a group member can, and can also get the invitation link and QR code. In the Should-have scope they can also remove members.
Creditor	A situational role of a group member who is owed money on a specific debt. They confirm or reject repayment reports.
Debtor	A situational role of a group member who owes money on a specific debt. They report full repayment of that debt.

6.3 Use Case List

ID	Use case	Primary actor	Priority	Related FR
UC-01	Register account	Visitor	Must	FR-01, FR-02
UC-02	Sign in	Visitor	Must	FR-03, FR-04
UC-03	Sign out	Signed-in user	Must	FR-07, FR-08
UC-04	Manage bank details	Signed-in user	Must	FR-05, FR-06
UC-05	Create group	Signed-in user	Must	FR-09, FR-10
UC-06	Share invitation link / QR code	Group organizer	Must	FR-11, FR-12
UC-07	Join group via invitation link	Signed-in user	Must	FR-13, FR-14, FR-17, FR-47
UC-08	View group members	Group member	Must	FR-15, FR-16
UC-09	Record expense	Group member	Must	FR-18, FR-19, FR-20, FR-21, FR-22, FR-23, FR-24

UC-10	View expense details	Group member	Must	FR-25
UC-11	Edit or delete expense	Creditor	Must	FR-26, FR-27, FR-28
UC-12	View debts	Group member	Must	FR-29, FR-30, FR-39
UC-13	Report repayment	Debtor	Must	FR-31, FR-32, FR-36, FR-38
UC-14	Confirm or reject repayment	Creditor	Must	FR-33, FR-34, FR-35, FR-37, FR-39
UC-15	Leave group	Group member	Should	FR-40, FR-42, FR-43, FR-44, FR-45
UC-16	Remove member	Group organizer	Should	FR-41, FR-42, FR-43, FR-44, FR-46
UC-17	Generate VietQR code	Debtor	Should	FR-48, FR-49
UC-18	View expense and repayment history	Group member	Could	FR-50, FR-51

7 Use Case Specifications

7.1 Use Case UC-01

Use case ID	UC-01
Use case name	Register account
Brief description	Allows a visitor to create a SplitNest account using valid registration information.
Primary actor	Visitor
Secondary actors	None
Priority	Must
Related requirements	FR-01, FR-02
Trigger	The visitor chooses to register an account.
Pre-conditions	1. The visitor is not signed in. 2. The registration page is accessible.
Post-conditions	Success: The account is created and the user is signed in. Failure: No account is created.
Main flow	1. The visitor opens the registration page. 2. The system displays the registration form. 3. The visitor enters the required registration information. 4. The visitor submits the registration form. 5. The system validates the submitted information. 6. The system checks whether the email address is already registered. 7. The system creates the account and securely stores the password. 8. The system signs in the newly registered user.

	9. The system redirects the user to the home screen.
Alternative flows	None
Exception flows	E1 (at step 6): Required information is missing or invalid. The system rejects the submission and displays appropriate validation errors. The visitor may correct the information and resubmit the form. E2 (at step 7): The email address is already registered. The system rejects the registration and informs the visitor that the email address is already in use. No new account is created.
Business rules	None
Non-functional constraints	NFR-03, NFR-04

7.2 Use Case UC-02

Use case ID	UC-02
Use case name	Sign in
Brief description	Allows a registered user to sign in to SplitNest using valid credentials.
Primary actor	Visitor
Secondary actors	None
Priority	Must
Related requirements	FR-03, FR-04
Trigger	The visitor chooses to sign in to SplitNest.
Pre-conditions	1. The visitor is not signed in. 2. The sign-in page is accessible.
Post-conditions	Success: The user is authenticated and can access features available to signed-in users. Failure: The visitor remains unauthenticated.
Main flow	1. The visitor opens the sign-in page. 2. The system displays the sign-in form.

	3. The visitor enters their registered email address and password. 4. The visitor submits the form. 5. The system validates the submitted information. 6. The system verifies the credentials against the registered account. 7. The system establishes an authenticated session. 8. The system redirects the user to the appropriate page and confirms successful sign-in.
Alternative flows	None
Exception flows	E1 (at step 5): Required information is missing or invalid. The system rejects the submission and displays appropriate validation errors. E2 (at step 6): The email address or password is incorrect. The system rejects the sign-in attempt and displays an error message without authenticating the visitor.
Business rules	None
Non-functional constraints	NFR-03, NFR-04

7.3 Use Case UC-03

Use case ID	UC-03
Use case name	Sign out
Brief description	Allows an authenticated user to sign out and end their session.
Primary actor	Signed-in user
Secondary actors	None
Priority	Must
Related requirements	FR-07, FR-08
Trigger	The user selects the sign-out option.
Pre-conditions	The user is signed in.

Post-conditions	Success: The user's session ends, and protected group data is no longer accessible until they sign in again.
Main flow	1. The user selects “Sign out”. 2. The system ends the user's session. 3. The system redirects the user to an appropriate public page.
Alternative flows	None
Exception flows	E1: If the sign-out request fails, the system informs the user that sign-out could not be completed. The session must not be treated as ended unless the system has actually invalidated it.
Business rules	None
Non-functional constraints	NFR-02, NFR-04, NFR-07.

7.4 Use Case UC-04

Use case ID	UC-04
Use case name	Manage bank details
Brief description	Allows a signed-in user to view and edit their own saved bank details.
Primary actor	Signed-in user
Secondary actors	None
Priority	Must
Related requirements	FR-05, FR-06
Trigger	The user opens their account settings or bank details section.
Pre-conditions	The user is signed in.
Post-conditions	Success: The user's bank details are saved and updated. Failure: The existing saved details remain unchanged.
Main flow	1. The user opens the bank details section.

	2. The system displays the user's saved bank name, account number, and account holder name, if available. 3. The user edits one or more fields. 4. The user selects “Save Changes”. 5. The system validates the submitted information. 6. The system saves the updated bank details as text. 7. The system confirms that the changes were saved.
Alternative flows	A1 (at step 2): If the user has not saved bank details, the system displays empty fields. A2 (at step 3): The user leaves the existing values unchanged and exits the section without saving.
Exception flows	E1 (at step 5): If the submitted information is invalid according to the system's validation rules, the system displays validation errors and does not save the invalid changes.
Business rules	BR-15
Non-functional constraints	NFR-02, NFR-04, NFR-07, NFR-10, NFR-12

7.5 Use Case UC-05

Use case ID	UC-05
Use case name	Create group
Brief description	Allows a signed-in user to create a private group for managing shared expenses.
Primary actor	Signed-in user
Secondary actors	None
Priority	Must
Related requirements	FR-09, FR-10
Trigger	The user chooses to create a group.
Pre-conditions	The user is signed in.

Post-conditions	Success: A new private group is created, and the creator is recorded as a member. Failure: No group is created.
Main flow	1. The user opens the group creation form. 2. The system displays the required group information fields. 3. The user enters a group name. 4. The user submits the form. 5. The system validates the submitted information. 6. The system creates the private group. 7. The system records the creator as a member of the group. 8. The system displays the created group and confirms success.
Alternative flows	None
Exception flows	E1 (at step 5): If the group name is empty or the submitted information is invalid, the system rejects the request and displays an appropriate error. E2 (at step 6): If the group cannot be created, the system reports the failure and does not display the group as successfully created.
Business rules	BR-03, BR-04
Non-functional constraints	NFR-02, NFR-04, NFR-07, NFR-08, NFR-09, NFR-10

7.6 Use Case UC-06

Use case ID	UC-06
Use case name	Share invitation link / QR code
Brief description	Allows a group organizer to retrieve the group's invitation link and display a QR code containing that link.
Primary actor	Group organizer
Secondary actors	None
Priority	Must

Related requirements	FR-11, FR-12
Trigger	The group organizer opens the group's invitation section.
Pre-conditions	The user is signed in and is the organizer of an existing group.
Post-conditions	Success: The invitation link is displayed, and a QR code containing the link is available.
Main flow	1. The organizer opens the selected group's invitation section. 2. The system verifies that the user is the group's organizer. 3. The system retrieves the group's invitation link. 4. The system displays the invitation link. 5. The system displays a QR code containing the invitation link.
Alternative flows	None
Exception flows	E1 (at step 2): If the user is not the organizer of the selected group, the system denies access to the organizer-only invitation function. E2 (at step 3): If the invitation link cannot be retrieved, the system reports the error instead of displaying an invalid link or QR code.
Business rules	BR-04
Non-functional constraints	NFR-02, NFR-04, NFR-07, NFR-09, NFR-10

7.7 Use Case UC-07

Use case ID	UC-07
Use case name	Join group via invitation link
Brief description	Allows a signed-in user to join a private group through a valid invitation link.
Primary actor	Signed-in user
Secondary actors	Group organizer
Priority	Must

Related requirements	FR-13, FR-14, FR-17, FR-47
Trigger	The user opens a group's invitation link.
Pre-conditions	The user is signed in and has an invitation link.
Post-conditions	Success: The user becomes a member of the group and can access its protected data. Failure: The user is not added to the group and gains no new access.
Main flow	1. The user opens the invitation link. 2. The system identifies the group associated with the link. 3. The system validates the invitation link. 4. The system checks whether the user is already a member. 5. The system adds the user to the group. 6. The system confirms successful membership and provides access to the group's permitted data.
Alternative flows	A1 (at step 4): If the user is already a member, the system does not create duplicate membership and may display the existing group. A2 (at step 5, former member rejoining): If the user previously left or was removed, the system may add them as a new member through the valid invitation link, as specified by FR-47.
Exception flows	E1 (at step 3): If the link is invalid, expired, or does not identify an existing group, the system rejects the request and displays an error. E2 (at step 5): If membership creation fails, the system reports the failure and does not grant group access.
Business rules	BR-03
Non-functional constraints	NFR-02, NFR-04, NFR-07, NFR-08, NFR-09, NFR-10

7.8 Use Case UC-08

Use case ID	UC-08
Use case name	View group members

Brief description	Allows a group member to view the member list of their group.
Primary actor	Group member
Secondary actors	None
Priority	Must
Related requirements	FR-15, FR-16
Trigger	The user opens the member list for a group.
Pre-conditions	The user is signed in and belongs to the selected group.
Post-conditions	Success: The group's member list is displayed. No group data is disclosed to a non-member.
Main flow	1. The user opens a group they belong to. 2. The user selects the member list. 3. The system verifies the user's membership in the group. 4. The system retrieves the group's member list. 5. The system displays the list to the user.
Alternative flows	None
Exception flows	E1 (at step 3): If the user does not belong to the group, the system denies access and returns no protected group information. E2 (at step 4): If the member list cannot be retrieved, the system displays an error.
Business rules	BR-04
Non-functional constraints	NFR-01, NFR-02, NFR-04, NFR-07, NFR-09, NFR-10, NFR-12

7.9 Use Case UC-09

Use case ID	UC-09
Use case name	Record expense

Brief description	Allows a group member to record a shared expense, split its amount equally among selected participants, and create repayment obligations.
Primary actor	Group member
Secondary actors	None
Priority	Must
Related requirements	FR-18, FR-19, FR-20, FR-21, FR-22, FR-23, FR-24
Trigger	The user chooses to add an expense to a group.
Pre-conditions	The user is signed in and belongs to the selected group.
Post-conditions	Success: The expense is recorded, participant shares sum to the total, and repayment obligations are created for participants other than the payer. Failure: The expense is not recorded.
Main flow	1. The user opens the expense creation form for a group. 2. The system verifies that the user is a member of the group. 3. The user enters the expense amount and other required expense information. 4. The user selects exactly one payer. 5. The user selects at least one participant who will share the expense. 6. The user submits the expense. 7. The system validates the amount, payer, participants, and group membership of each selected person. 8. The system divides the amount equally among the selected participants in whole VND, distributing any remainder in fixed member-ID order. 9. The system verifies that the sum of the allocated shares equals the expense amount. 10. The system records the expense and its allocated shares. 11. The system creates an individual repayment obligation for each participant other than the payer, linked to the source expense. 12. The system confirms that the expense was recorded.

Alternative flows	A1 (at step 8): If the amount cannot be divided evenly, the system distributes the remaining VND according to the fixed member-ID ordering rule. A2 (at step 11): If the payer is also a participant, the system does not create a repayment obligation for the payer.
Exception flows	E1 (at step 7): If the amount is zero, negative, non-integer, or otherwise invalid, the system rejects the expense and displays validation errors. E2 (at step 7): If no participant is selected, more than one payer is selected, or a payer or participant does not belong to the group, the system rejects the expense. E3 (at step 2): If the user is not a member of the group, the system denies access and does not record the expense. E4 (at step 9): If the allocated shares do not sum to the total amount, the system must not save the inconsistent expense.
Business rules	BR-01, BR-04 → BR-07
Non-functional constraints	NFR-01, NFR-02, NFR-04, NFR-05, NFR-07, NFR-08, NFR-09, NFR-10

7.10 Use Case UC-10

Use case ID	UC-10
Use case name	View expense details
Brief description	Allows a group member to view an expense's details, including its payer, participants, and allocated amounts.
Primary actor	Group member
Secondary actors	None
Priority	Must
Related requirements	FR-25
Trigger	The user selects an expense to view its details.
Pre-conditions	The user is signed in, belongs to the group containing the expense, and the expense exists.
Post-conditions	Success: The expense details are displayed to the authorized user. No protected expense information is disclosed to a non-member.

Main flow	1. The user opens the relevant group. 2. The user selects an expense. 3. The system verifies that the user is a member of the group. 4. The system retrieves the expense details. 5. The system displays the expense amount, payer, participants, and allocated amounts.
Alternative flows	None
Exception flows	E1 (at step 3): If the user is not a member of the group, the system denies access and returns no protected expense information. E2 (at step 4): If the expense does not exist or cannot be retrieved, the system displays an appropriate error.
Business rules	BR-04 → BR-06
Non-functional constraints	NFR-01, NFR-02, NFR-04, NFR-05, NFR-07, NFR-09, NFR-10, NFR-12

7.11 Use Case UC-11

Use case ID	UC-11
Use case name	Edit or delete expense
Brief description	Allows the member who created an expense to edit or delete it before any related repayment report has been submitted.
Primary actor	Group member who created the expense
Secondary actors	None
Priority	Must
Related requirements	FR-26, FR-27, FR-28
Trigger	The expense creator selects the option to edit or delete an expense.
Pre-conditions	The user is signed in, belongs to the group, created the expense, and no related repayment report has been submitted.

Post-conditions	Success: The expense is updated or deleted, and its related debts are recalculated or removed as appropriate. Failure: The expense and related records remain unchanged.
Main flow	1. The expense creator opens the expense details. 2. The system verifies the user's group membership and ownership of the expense. 3. The system checks whether any related repayment report has been submitted. 4. The user chooses to edit or delete the expense. 5. If editing: The system displays the editable expense information, and the user makes changes and submits them. 6. If editing: The system validates the updated expense and recalculates the related shares and debts. 7. If deleting: The system processes the deletion and removes the related debts as appropriate. 8. The system saves the changes and confirms the result.
Alternative flows	A1 (at step 4): The user chooses to cancel and leaves the expense unchanged. A2 (at step 4): The user chooses either editing or deletion; only the selected operation is performed.
Exception flows	E1 (at step 2): If the user did not create the expense, the system denies the operation. E2 (at step 3): If any related repayment report exists, including a rejected report, the system rejects both editing and deletion and explains why. E3 (at step 6): If the edited expense violates the expense validation rules, the system rejects the changes and preserves the existing expense.
Business rules	BR-08
Non-functional constraints	NFR-01, NFR-02, NFR-04, NFR-05, NFR-06, NFR-07, NFR-09, NFR-10

7.12 Use Case UC-12

Use case ID	UC-12
Use case name	View debts

Brief description	Allows a group member to view the amounts they owe and are owed within a group.
Primary actor	Group member
Secondary actors	None
Priority	Must
Related requirements	FR-29, FR-30, FR-39
Trigger	The user opens the group's debt view.
Pre-conditions	The user is signed in and belongs to the selected group.
Post-conditions	Success: The system displays the user's debts and amounts owed, reflecting the current repayment statuses.
Main flow	1. The user opens the selected group. 2. The user opens the debt view. 3. The system verifies the user's membership in the group. 4. The system retrieves the individual repayment obligations in the group. 5. The system calculates the amounts the user owes and is owed without offsetting obligations in opposite directions. 6. The system accounts for repayment decisions: confirmed obligations are excluded from outstanding totals, while rejected obligations remain outstanding and pending reports remain unsettled. 7. The system displays the debt information to the user.
Alternative flows	A1 (at step 4): If the group has no repayment obligations, the system displays an empty debt view or indicates that no debts have been recorded.
Exception flows	E1 (at step 3): If the user is not a member of the group, the system denies access and returns no protected debt information. E2 (at step 4): If the debt information cannot be retrieved, the system reports an error rather than displaying an unverified total.
Business rules	BR-04, BR-09, BR-11 → BR-14
Non-functional constraints	NFR-01, NFR-02, NFR-04, NFR-07, NFR-08, NFR-09, NFR-10, NFR-12

7.13 Use Case UC-13

Use case ID	UC-13
Use case name	Report repayment
Brief description	Allows a debtor to report full repayment of an outstanding obligation made outside SplitNest and submit for the creditor's confirmation
Primary actor	Debtor
Secondary actors	None
Priority	Must
Related requirements	FR-31, FR-32, FR-36, FR-38
Trigger	The debtor chooses to report repayment of an outstanding obligation.
Pre-conditions	1. The user is signed in and belongs to the selected group. 2. The user is the debtor of an existing obligation with status Outstanding.
Post-conditions	Success: A repayment report is recorded, and the obligation changes to Pending Confirmation while remaining unsettled. Failure: No repayment report is created, and the obligation remains unchanged.
Main flow	1. The debtor opens the selected group's debt view. 2. The debtor selects an outstanding obligation they owe. 3. The system verifies the user's group membership and that they are the debtor of the selected obligation. 4. The system retrieves the obligation's full amount and current status. 5. The debtor selects "Mark as paid". 6. The system displays the full repayment amount and requests confirmation that the repayment was made outside the application. 7. The debtor confirms that the full amount has been repaid.

	8. The system verifies that the obligation is still Outstanding and has no pending repayment report. 9. The system records the repayment report and changes the obligation status to Pending Confirmation. 10. The system confirms that the report was submitted and is awaiting the creditor's decision.
Alternative flows	A1 (at step 7): If the debtor cancels the repayment report, the system returns to the debt view without creating a report or changing the obligation status.
Exception flows	E1 (at step 3): If the user does not belong to the group or is not the debtor of the selected obligation, the system denies the operation. E2 (at step 7): If a submitted repayment report covers only part of the obligation, the system rejects the report because partial repayments are not supported. E3 (at step 8): If the obligation is already Pending Confirmation or Confirmed, the system rejects the request and does not create another pending report. E4 (at step 9): If the repayment report cannot be saved, the system reports the failure and does not change the obligation status.
Business rules	BR-02, BR-04, BR-10, BR-11, BR-13
Non-functional constraints	NFR-01, NFR-02, NFR-04, NFR-06, NFR-07, NFR-09, NFR-10

7.14 Use Case UC-14

Use case ID	UC-14
Use case name	Confirm or reject repayment
Brief description	Allows the creditor of a repayment obligation to confirm or reject a pending repayment report, updating the obligation status and outstanding debt totals accordingly.
Primary actor	Creditor

Secondary actors	None
Priority	Must
Related requirements	FR-33, FR-34, FR-35, FR-37, FR-39
Trigger	The creditor selects a pending repayment report to confirm or reject.
Pre-conditions	1. The user is signed in and belongs to the selected group. 2. The user is the creditor of the selected obligation. 3. The obligation has a repayment report with status Pending Confirmation.
Post-conditions	Success (confirmation): The repayment is confirmed, the obligation changes to Confirmed, and its amount is excluded from outstanding debt totals. Success (rejection): The repayment report is rejected, the obligation returns to Outstanding, and its amount remains outstanding. Failure: No repayment decision is recorded, and the existing obligation status remains unchanged.
Main flow	1. The creditor opens the selected group's debt view. 2. The creditor selects a repayment report awaiting confirmation. 3. The system verifies the user's group membership and that they are the creditor of the selected obligation. 4. The system retrieves the repayment report and verifies that it is Pending Confirmation. 5. The system displays the obligation details, repayment amount, and available confirmation options. 6. The creditor selects "Confirm Received". 7. The system verifies that the repayment report is still pending and that the obligation has not already been confirmed. 8. The system records the confirmation and changes the obligation status to Confirmed. 9. The system updates the displayed debt totals, excluding the confirmed obligation from outstanding amounts.

	10. The system confirms that the repayment was accepted and the obligation was settled.
Alternative flows	A1 (at step 6): If the creditor selects "Reject / Not Received", the system verifies that the report is still pending, records the rejection, and returns the obligation status to Outstanding. The debt remains included in outstanding totals, and the system confirms that the repayment report was rejected. A2 (at step 6): If the creditor cancels the operation, the system returns to the debt view without recording a decision or changing the obligation status.
Exception flows	E1 (at step 3): If the user does not belong to the group or is not the creditor of the selected obligation, the system denies the operation. E2 (at step 4): If the repayment report does not exist or cannot be retrieved, the system displays an appropriate error. E3 (at step 7): If the report is no longer Pending Confirmation or the obligation has already been confirmed, the system rejects the decision and prevents duplicate settlement. E4 (at step 8): If the decision cannot be saved, the system reports the failure and does not treat the repayment as confirmed or rejected.
Business rules	BR-02, BR-04, BR-11 → BR-14
Non-functional constraints	NFR-01, NFR-02, NFR-04, NFR-06, NFR-07, NFR-09, NFR-10

8 Prototype / Mockups

Accessibility: Contrast and Labels are ensured in the mockups. Because the Figma prototype is not a working application, its text fields cannot accept typed input and keyboard navigation. Keyboard support per NFR-10 will be implemented in the implementation and verified in the accessibility audit..

Screen	Related use case(s)	Mockup / link
Account Screen	UC-01: Register Account, UC-02: Sign In, UC-03: Sign Out, UC-04: Manage Bank Details	Account Screen
Group Screen	UC-05: Create Group, UC-06: Share Invitation Link / QR Code, UC-07: Join Group via Invitation Link, UC-08: View Group Members, UC-15: Leave Group, UC-16: Remove Member	Group Screen
Expense Screen	UC-09: Record Expense, UC-10: View Expense Details, UC-11: Edit or Delete Expense	Expense Screen
Debt Screen	UC-12: View Debts, UC-13: Report Repayment, UC-14: Confirm or Reject Repayment	Debt Screen
VietQR Screen	UC-17: Generate VietQR Code	VietQR Screen
History Screen	UC-18: View Expense and Repayment History	History Screen

9 Requirements Traceability Matrix

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
FR-01	UC-01	Account Screen		
FR-02	UC-01	Account Screen		
FR-03	UC-02	Account Screen		
FR-04	UC-02	Account Screen		
FR-05	UC-04	Account Screen		
FR-06	UC-04	Account Screen		
FR-07	UC-03	Account Screen		
FR-08	UC-03	Account Screen		
FR-09	UC-05	Group Screen		
FR-10	UC-05	Group Screen		
FR-11	UC-06	Group Screen		
FR-12	UC-06	Group Screen		
FR-13	UC-07	Group Screen		
FR-14	UC-07	Group Screen		
FR-15	UC-08	Group Screen		
FR-16	UC-08	Group Screen		
FR-17	UC-07	Group Screen		
FR-18	UC-09	Expense Screen		
FR-19	UC-09	Expense Screen		

FR-20	UC-09	Expense Screen		
FR-21	UC-09	Expense Screen		
FR-22	UC-09	Expense Screen		
FR-23	UC-09	Expense Screen		
FR-24	UC-09	Expense Screen		
FR-25	UC-10	Expense Screen		
FR-26	UC-11	Expense Screen		
FR-27	UC-11	Expense Screen		
FR-28	UC-11	Expense Screen		
FR-29	UC-12	Debt Screen		
FR-30	UC-12	Debt Screen		
FR-31	UC-13	Debt Screen		
FR-32	UC-13	Debt Screen		
FR-33	UC-14	Debt Screen		
FR-34	UC-14	Debt Screen		
FR-35	UC-14	Debt Screen		
FR-36	UC-13	Debt Screen		
FR-37	UC-14	Debt Screen		
FR-38	UC-13	Debt Screen		
FR-39	UC-12, UC-14	Debt Screen		
FR-40	UC-15	Group Screen		
FR-41	UC-16	Group Screen		
FR-42	UC-15, UC-16	Group Screen		

FR-43	UC-15, UC-16	Group Screen		
FR-44	UC-15, UC-16	Group Screen		
FR-45	UC-15	Group Screen		
FR-46	UC-16	Group Screen		
FR-47	UC-07	Group Screen		
FR-48	UC-17	VietQR Screen		
FR-49	UC-17	VietQR Screen		
FR-50	UC-18	History Screen		
FR-51	UC-18	History Screen		
FR-52	N/A	N/A		
FR-53	N/A	N/A		

Notice: Separate table for the non-functional requirements as some rows do not apply for them.

Req. ID	Category	Use case	Verification	Test cases (PA4)
NFR-01	Performance	UC-09 → UC-14	Load testing; measure P95	
NFR-02	Security	UC-08 → UC-14	API authorization tests	
NFR-03	Security	UC-01, UC-02	Check database and logs	
NFR-04	Security	All	Check deployed protocols	
NFR-05	Data integrity	UC-09, UC-10	Unit tests for the splitting algorithm	
NFR-06	Concurrency	UC-14	Concurrent integration tests	
NFR-07	Reliability	All	Persistence tests	
NFR-08	Usability	UC-05, UC-07, UC-09, UC-12	Usability testing	

NFR-09	Compatibility	UC-01 → UC-14	Browser testing	
NFR-10	Accessibility	UC-01 → UC-14	Accessibility audit	
NFR-11	Maintainability	N/A	GitHub Actions/branch protection	
NFR-12	Privacy	UC-08, UC-10, UC-12, UC-17, UC-18	API response inspection	

10 Requirements Review

Participants:

24127455 - Lại Phan Nhật Nam

24127295 - Phạm Hoàng Khôi Nguyên

24127308 - Nguyễn Hoàng Tiến

24127459 - Trần Văn Nam

24127409 - Ngô Phúc Khang

Issue Date: 02/11/2026

#	Issue found	Requirement(s)	Resolution
1	Sign-out functionality was not explicitly specified in the original requirements.	FR-07, FR-08; UC-03	The team reviewed whether sign-out should be included in the first release. Its inclusion was retained as a scope decision, with session termination specified to protect group data after sign-out.
2	Leaving a group and removing a member were not explicitly specified in the original requirements.	FR-40–FR-46; UC-15, UC-16	The team reviewed these functions as additional requirements. Their inclusion was treated as a scope decision and assigned Should-have priority. The team should confirm that these additions are acceptable within the agreed project scope.
3	The phrase "private bank details" in the previous version of NFR-12 was ambiguous and did not identify which information required protection or the conditions under which it could be accessed.	NFR-12	The requirement was rewritten to specify the handling of saved bank details in group, expense and debt list APIs, including the conditions under which a debtor may access a creditor's bank details. The verification method remains API response inspection.
4	FR-47 was accidentally omitted from the use case list, leaving the requirement without an explicit use case association.	FR-47; UC-07	The omission was corrected by associating FR-47 with UC-07 (Join group via invitation link), as rejoining a group uses the existing invitation workflow. The use case list and traceability matrix were checked for consistency.

11 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	24127459	ChatGPT	Drafted the Purpose and Scope section.	Checked against PA0 and PA1.
2	24127459	ChatGPT	Identified actors and use cases for the Use Case Model.	Compared with PA1 requirements and corrected relationships.
3	24127459	ChatGPT	Mapped prototype screens to related use cases.	Checked against the assigned mockup tasks in PA1.
4	24127459	OpenCode (Kimi)	Review PA0–PA2 and redesign Account, Group/Invite, and Expense mockups.	Checked requirements, links, and Chrome rendering.
5	24127459	OpenCode (Kimi)	Improve the mockups with a friendly, cute style.	Checked design, responsiveness, and text contrast.
6	24127459	OpenCode (Kimi)	Update documentation and export PNG files.	Reviewed documentation and inspected all PNGs.
7	24127295	ChatGPT	Refined the PA1 Vision for PA2.	Checked consistency with PA0/PA1 and clarified expense rules.
8	24127295	ChatGPT	Drafted functional requirements for F01–F08.	Checked feature IDs, priorities and business rules against PA1.
9	24127295	ChatGPT	Drafted measurable non-functional requirements.	Reviewed proposed metrics and verification methods.
10	24127295	ChatGPT	Reviewed API-related NFRs and scope limitations.	Checked against PA1 architecture and excluded features.
11	24127295	ChatGPT	Translated requirement tables into English.	Preserved the original Word document formatting.
12	24127295	ChatGPT	Reviewed AI usage logs and Git evidence.	Checked consistency with repository documentation.

13	24127409	Gemini	Draft acceptance criteria scenarios for Section 5.4.	Verified scenarios followed the Given/When/Then structure.
14	24127409	Gemini	Synthesize the Use Case List and Actor roles for Section 6.2 and 6.3.	Checked use cases, actor hierarchy and FR mappings against PA1.
15	24127409	Gemini	Outline the preliminary Master Test Plan structure in preparation for PA4.	Outlined the Master Test Plan for PA4.
16	24127455	Claude	Explained how to build the Requirements Traceability Matrix, with step-by-step examples.	Compared against Sections 5 and 6.3 of the SRS. I built the FR-to-UC mapping myself and checked it for 53 FR rows with no duplicates.
17	24127455	Claude	Advised whether NFRs belong in the RTM and how to lay them out.	Choose a separate NFR table. Checked each NFR-to-UC mapping against Section 5.2.
18	24127455	Claude	Reviewed the SRS for consistency and flagged problems (UC numbering, FR-47 with no use case, AC-04.1 vs NFR-12, missing ACs).	Checked each flagged point against the document. Kept the valid ones and logged them in Section 10.
19	24127455	Claude	Reviewed the SRS for major, minor and potential contradictions.	Checked each flagged item against the cited FR/BR/NFR/AC text. Corrected one finding: the "keyboard support is missing" note refers to Figma prototype text fields that can't take input, not a product gap.
20	24127308	Claude	Map each use case to related business rules	Compared the references to the official list to make sure they are correct.
21	24127308	ChatGPT	Draft Problem Statement for PA2 using the provided documents.	Compared with PA1 Preliminary Problem Statement.
22	24127308	Claude	Check for inconsistencies in Problem Statement	Proofread against current PA2 version.
23	24127295	ChatGPT	Requested draft for section 7	Checked proposed actors, FR/BR/NFR references and flows against Sections 5, 6.